MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER SWITCHING
TYPES 2N5664, 2N5665, 2N5666, AND 2N5667
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors for use in high-speed power switching applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 (TO-66) and 2 (TO-5).

1.3 Maximum ratings.

Type	P_T T_A = +25°C	P_T T_C = +100°C	V_{EBO}	V_{CBO}	V_{CEO}	I_C	I_B	T_{STG} and T_{OP}
	W	W	V dc	V dc	V dc	A dc	A dc	°C
						5	1	-65 to +200
2N5664	2.5 1/	30 3/	6	250	200			
2N5665	2.5 1/	30 3/	6	400	300			
2N5666	1.2 2/	15 4/	6	250	200			
2N5667	1.2 2/	15 4/	6	400	300			

1/ Derate linearly 14.3 mW/°C for T_A > +25°C.
2/ Derate linearly 6.9 mW/°C for T_A > +25°C.
3/ Derate linearly 300 mW/°C for T_C > +100°C.
4/ Derate linearly 150 mW/°C for T_C > +100°C.

1.4 Primary electrical characteristics at T_A = +25°C.

Limits	h_{FE} V_{CE} = 5 V dc I_C = 1 A dc		h_{fe} V_{CE} = 5 V dc I_C = 0.5 A dc f = 10 MHz	$V_{BE(sat)}$ I_C = 3 A dc 1/	$V_{CE(sat)}$ I_C = 3 A dc 1/	Pulse response		
						t_{on} I_C = 1 A dc	t_{off} I_C = 1 A dc	
	2N5665 2N5667	2N5664 2N5666					2N5664 2N5666	2N5665 2N5667
				V dc	V dc	µs	µs	µs
Min	25	40	2.0					
Max	75	120	7.0	1.2	0.4	0.25	1.5	2.0

1/ I_B = 0.3 A dc for 2N5664, 2N5666; I_B = 0.6 A dc for 2N5665, 2N5667.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rome Air Development Center, ATTN: RBE-2, Griffiss AFB, NY 13441, using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type	$R_{\theta JC}$
	°C/W (max)
2N5664, 2N5665	3.3
2N5666, 2N5667	6.7

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and as shown on figures 1, 2, and 3 herein.

3.3.1 Lead material and finish. The lead material for types 2N5666 and 2N5667 shall be Kovar or Alloy 52. The lead finish shall be gold- or tin-plated. The lead finish may be specified in the contract or purchase order (see 6.2) without affecting the qualified product status of the device or applicable JAN marking.

3.4 Marking. Device marking for device types 2N5664 and 2N5665 shall be in accordance with MIL-S-19500. For device types 2N5666 and 2N5667 the following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS levels	JANTX and JANTXV levels
9	I_{CES1} and h_{FE2}	I_{CES1}
11	ΔI_{CES1} and h_{FE2}; ΔI_{CES} =100% of initial value or 10 nA dc, whichever is greater; h_{FE2} =±15% of initial value.	I_{CES1} and h_{FE2}; ΔI_{CES1} =100% of initial value or 20 nA dc, whichever is greater.
12	See 4.2.1	See 4.2.1
13	Subgroup 2 of table I herein; ΔI_{CES1} = 100% of initial value or 10 nA dc, whichever is greater; Δh_{FE2} = ±15% of initial value.	Subgroup 2 of table I herein; ΔI_{CES1} = 100% of initial value or 20 nA dc, whichever is greater; Δh_{FE2} = ±25% of initial value.

4.2.1 Power burn-in test conditions. Power burn-in test conditions are as follows:

T_J = +187.5 ±12.5°C V_{CE} = 100 V dc $T_A \leq$ +100°C

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

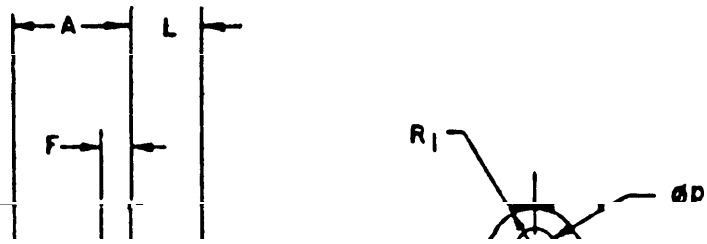
4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

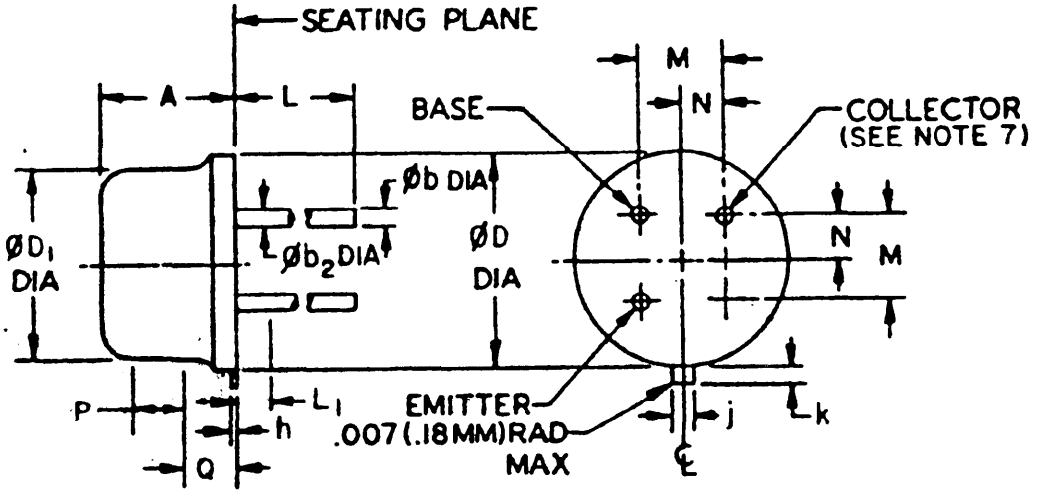
4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and table IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

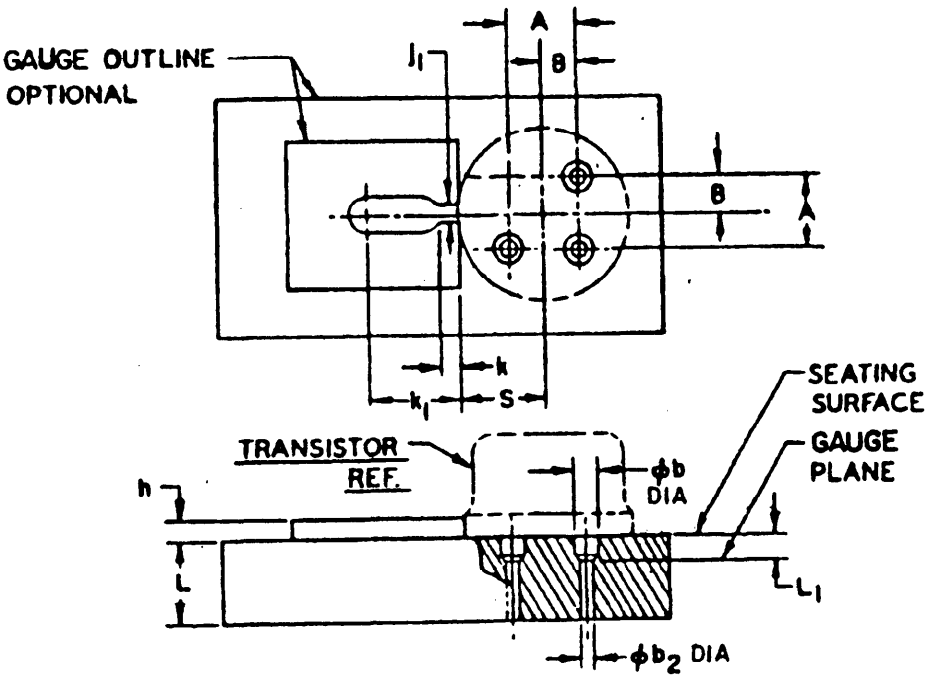

A
L
F
R1
ØD



DIMENSIONS					NOTES
	INCHES		MILLIMETERS		
LTR	MIN	MAX	MIN	MAX	
ØD	.335	.370	8.51	9.40	
ØD1	.305	.335	7.75	8.51	
A	.240	.260	6.10	6.60	
L	1.500	1.750	38.10	44.45	9
L1		.050		1.27	10
Øb	.016	.021	.41	.53	2, 9
Øb2	.016	.019	.41	.48	3, 9
P	.100		2.54		4
Q					5
k	.029	.045	.74	1.14	8
j	.028	.034	.71	.86	
h	.009	.125	.23	3.18	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on dimension $ØD_1$ in this zone shall shall not exceed .010 (0.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gauging plane .054 + .001, -.000 (1.37 +.03, -00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 3 shows the preferred measured method.
7. The collector shall be electrically connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All three leads (see 3.3.1).
10. Diameter of leads in this zone is not controlled.

FIGURE 2. Physical dimensions of transistor types 2N5666 and 2N5667.



DIMENSIONS				
	INCHES		MILLIMETERS	
LTR	MIN	MAX	MIN	MAX
A	.1409	.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
S	.182	.199	4.62	5.05
k	.009	.011	.23	.28
k_1	.125 Nom		3.18 Nom	
L_1	.054	.055	1.37	1.40
L	.372	.378	9.45	9.60
j_1	.0350	.0355	.89	.90
h	.150 Nom		3.81 Nom	
$Øb_2$	.0325	.0335	.83	.85
$Øb_1$	.0595	.0605	1.51	1.54

NOTES:
1. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ±.50 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge.
2. The location of the tab locator, within the limits of dimension S, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents are given for general information only.

FIGURE 3. Gauge for lead and tab location for transistor types 2N5666 AND 2N5667.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.833 A dc for device types 2N5664 and 2N5665 and 0.41 A dc for device types 2N5666 and 2N5667.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25^{\circ}C \leq T_R \leq +75^{\circ}C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 3.3°C/W (2N5664, 2N5665) and 6.7°C/W (2N5666, 2N5667).

4.5.3 Group C life test. If the option of "T_C" life test is conducted in group B inspection, the test sample may be continued as subgroup 6 of group C inspection in accordance with MIL-S-19500 and table III herein.

4.5.4 Inspection conditions. Unless otherwise specified in MIL-S-19500 or herein, all inspections shall be conducted at a case temperature (T_C) of $+25^{\circ}C \pm 3^{\circ}C$.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. The contract or purchase order should specify lead finish if other than specified herein (see 3.3.1).

TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1							
Visual and mechanical examination	2071						
Subgroup 2							
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)		$V_{(BR)CEO}$			
2N5664, 2N5666					200		V dc
2N5665, 2N5667					300		V dc
Breakdown voltage, collector to emitter	3011	Bias condition B; I_C = 10 mA dc R_1 = 100Ω; pulsed (see 4.5.1)		$V_{(BR)CER}$			
2N5664, 2N5666					250		V dc
2N5665, 2N5667					400		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc		$V_{(BR)EBO}$	6		V dc
Collector to emitter cutoff current	3041	Bias condition C		I_{CES1}		0.2	µA dc
2N5664, 2N5666		V_{CE} = 200 V dc					
2N5665, 2N5667		V_{CE} = 300 V dc					
Collector to base cutoff current	3036	Bias condition D		I_{CBO}			
2N5664, 2N5666		V_{CB} = 200 V dc				0.1	µA dc
		V_{CB} = 250 V dc				1.0	mA dc
2N5665, 2N5667		V_{CB} = 300 V dc				0.1	µA dc
		V_{CB} = 400 V dc				1.0	mA dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)		h_{FE1}			
2N5664, 2N5666					40		
2N5665, 2N5667					25		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)		h_{FE2}			
2N5664, 2N5666					40	120	
2N5665, 2N5667					25	75	

See footnotes at end of table.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 2 - Continued							
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)		h_{FE3}			
2N5664, 2N5666					15		
2N5665, 2N5667					10		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5 A dc; pulsed (see 4.5.1)		h_{FE4}	5		
Collector-emitter voltage (saturated)	3071	I_C = 3.0 A dc		$V_{CE(sat)1}$		0.4	V dc
2N5664, 2N5666		I_B = 0.3 A dc; pulsed (see 4.5.1)					
2N5665, 2N5667		I_B = 0.6 A dc; pulsed (see 4.5.1)					
Collector-emitter voltage (saturated)	3071	I_C = 5 A dc; I_B = 1 A dc; pulsed (see 4.5.1)		$V_{CE(sat)2}$ 2/		1.0	V dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 3.0 A dc		$V_{BE(sat)1}$ 2/		1.2	V dc
2N5664, 2N5666		I_B = 0.3 A dc; pulsed (see 4.5.1)					
2N5665, 2N5667		I_B = 0.6 A dc; pulsed (see 4.5.1)					
Base emitter voltage (saturated)	3066	Test condition A; I_C = 5 A dc; I_B = 1 A dc; pulsed (see 4.5.1)		$V_{BE(sat)2}$		1.5	V dc
Subgroup 3							
High-temperature operation:		T_A = +150°C					
Collector to emitter cutoff current	3041	Bias condition C		I_{CES2}		100	µA dc
2N5664, 2N5666		V_{CE} = 200 V					
2N5665, 2N5667		V_{CE} = 300 V					
Low-temperature operation:		T_A = -55°C					
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)		h_{FE5}			
2N5664, 2N5666					15		
2N5665, 2N5667					10		

See footnotes at end of table.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 4							
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	I_C = 0.5 A dc; V_{CE} = 5 V dc; f = 10 MHz		$\|h_{fe}\|$	2.0	7.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; 100 kHz ≤ f ≤ 1 MHz		C_{obo}		120	pF
Pulse response:							
Turn-on time	3251	Test condition A; I_C = 1.0 A dc; V_{CC} = 100 V dc		t_{on}		0.25	µs
2N5664, 2N5666		See figure 4					
2N5665, 2N5667		See figure 5					
Turn-off time	3251	Test condition A; I_C = 1.0 A dc; V_{CC} = 100 V dc		t_{off}			
2N5664, 2N5666		See figure 4				1.5	µs
2N5665, 2N5667		See figure 5				2.0	µs
Subgroup 5							
Safe operating area (continuous dc) (for types 2N5664, 2N5665 only)	3051	T_C = +100°C; t ≥ 1 s, 1 cycle; $t_r + t_f$ = 10 µs (see figure 6)					
Test 1							
2N5664, 2N5665		V_{CE} = 6 V dc; I_C = 5 A dc					
Test 2							
2N5664, 2N5665		V_{CE} = 40 V dc; I_C = 0.75 A dc					
Test 3							
2N5664		V_{CE} = 200 V dc; I_C = 43 mA dc					
Test 4							
2N5665		V_{CE} = 300 V dc; I_C = 21 mA dc					

See footnotes at end of table.

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 5 -Continued							
Safe operating area (continuous dc)(for types 2N5666 and 2N5667 only)	3051	T_C = +100°C; $t \leq 1$ s, 1 cycle; $t_r + t_f$ = 10 µs (see figure 7)					
Test 1							
2N5666, 2N5667		V_{CE} = 3.0 V dc; I_C = 5 A dc					
Test 2							
2N5666, 2N5667		V_{CE} = 37.5 V dc; I_C = 0.4 A dc					
Test 3							
2N5666		V_{CE} = 200 V dc; I_C = 27 mA dc					
Test 4							
2N5667		V_{CE} = 300 V dc; I_C = 14 mA dc					
Safe operating area (switching)	3053	Load condition B (clamped inductive load) (see figure 8); T_C = +100°C; $t_r + t_f \leq 10$ µs duty cycle ≤ 2%; t_p 4 ms; R_S = 0.5 ohms; R_{BB1} = 50 ohms; V_{BB1} = 50 V dc; R_{BB2} = 50 ohms; V_{BB2} = -4 V dc; I_C = 5 A dc; V_{CC} = 50 V dc; $R_L \leq 2.5$ ohms; L = 40 mH (Triad C-48U or equivalent)					
2N5664, 2N5666		Clamp voltage = 200 +0, -5 V dc					
2N5665, 2N5667		Clamp voltage = 300 +0, -5 V dc					
Electrical measurements		See table IV, steps 1 and 4					
Subgroups 6 and 7							
Not applicable							

1/ For LTPD numbers, see MIL-S-19500.
2/ Measured at less than 0.125 inch from case.

TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figures 1 and 2		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
Fine leak				
Gross leak				
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Decap-internal visual (design verification)	2075			
SEM	2077			
Bond strength	2037	Test condition A		
Die shear (excluding axial leaded devices)	2017			
Subgroup 4				
Intermittent operation life	1037	V_{CB} = 30 V dc; P_T = 2.5 W at T_A = +25°C ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink nor forced air cooling on the devices shall be permitted.		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Subgroup 5				
Accelerated steady-state operation life	1027	V_{CB} = 30 V dc; T_A = +125°C ±25°C for 96 hours; P_T = 2.5 W at T_A = 112°C or adjusted as required by chosen T_A to give an average lot. T_J = +275°C. Marking legibility requirement shall not apply		

See footnote at end of table.

TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5 - Continued				
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Bond strength (Al-Au die interconnects only)	2037	Test condition A; all internal wires for each device shall be pulled separately.		
Subgroup 6				
Thermal resistance (forward voltage drop, emitter to base diode method).	3131	See 4.5.2		

1/ For LTPD numbers see MIL-S-19500.

TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
Fine leak			
Gross leak			
Electrical measurements		See table IV, steps 1 and 3	
Subgroup 3			
Steady-state operation life	1027	T_J = +187.5°C ±12.5°C; V_{CE} = 100 ±5 V dc; $T_A \leq$ +100°C	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, and 6	
Subgroup 4			
Decap internal visual design verification	2075		
Bond strength	2037	Test condition A	
Subgroup 5			
Not applicable			
Subgroup 6			
High-temperature life (non-operating)	1032	T_A = +200°C	
Electrical measurements		See table IV, steps 1, 4, 5, and 6	
Subgroup 7			
Safe operating area (unclamped inductive)	3053	Load condition C (unclamped inductive load) (see figure 9); T_C = +100°C; duty cycle ≤ 2%; $t_r + t_f \leq 10$ µs; R_S = 0.12 ohms; t_p 50 µs (vary to obtain I_C): R_{BB1} = 50 ohms; V_{BB1} = 50 V dc; R_{BB2} = 100 ohms; V_{BB2} = 0.5 V dc; I_C = 5 A dc; V_{CC} = 12.5 V dc; R = .033 ohms; L = .065 mH (20 turns 1 row of no. 20 ASG wire on Arnold Engineering Co. core no. A-930157-2, or equivalent)	
Electrical measurements		See table IV, steps 1 and 4	

TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1			15				
Physical dimensions	2066	(See figures 1 and 2)					
Subgroup 2			10				
Thermal shock (glass strain)	1056	Test condition B					
Terminal strength (tension) (2N5664 and 2N5665 only)	2036	Test condition A; weight = 3 lbs; t4me = 15 s					
Terminal strength (lead fatigue) (for 2N5666 and 2N5667 only)	2036	Test condition E					
Hermetic seal	1071						
Fine leak							
Gross leak							
Moisture resistance	1021						
Visual and mechanical examination	2071						
Electrical measurements		See table IV, steps 1, 2, and 4					
Subgroup 3			10				
Shock	2016						
Vibration, variable frequency	2056						
Electrical measurements		See table IV, steps 1, 2, and 4					
Subgroup 4			15				
Salt atmosphere (corrosion)	1041						
Subgroup 5			15				
Barometric pressure (reduced)	1001	Normal mounting; pressure = 8 mmHg for 60 s minimum					
Measurement during test:							
Collector to base cutoff current	3036	Bias condition D		I_{CBO}			
2N5664, 2N5666		V_{CB} = 250 V dc				1	mA dc
2N5665, 2N5667		V_{CB} = 400 V dc				1	mA dc

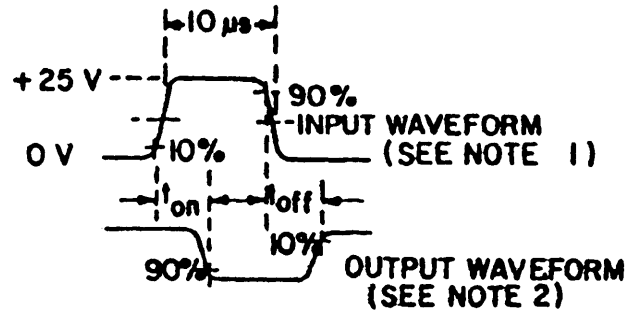
TABLE III. Group C inspection (all quality levels) - Continued.

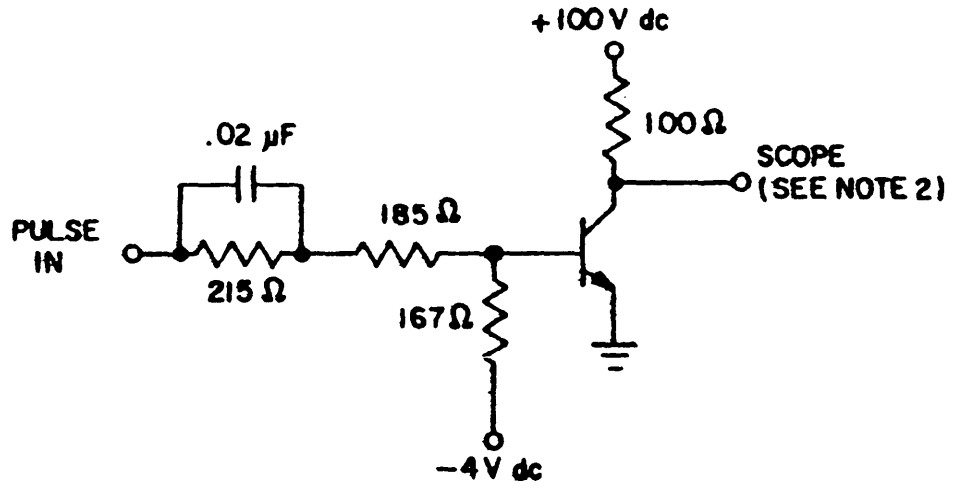
Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 6			10				
Steady-state operation life	1027						
2N5664, 2N5666		T_C = +100°C; P_T = 30 W; V_{CE} = 30 V dc					
2N5665, 2N5667		T_A = +25°C; P_T = 1.2 W; V_{CE} = 40 V dc					
End-point electrical measurements		See table IV, steps 1, 2, 3, 4, 5, and 6					
Subgroup 7			15				
Thermal resistance	3131	See 4.5.2				See 4.5.2	

TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}			
	2N5664, 2N5666		V_{CE} = 200 V dc			0.2	µA dc
	2N5665, 2N5667		V_{CE} = 300 V dc			0.2	µA dc
2.	Collector to emitter voltage (saturated)	3071	I_C = 5 A dc; I_B = 1 A dc (pulsed, see 4.5.1) 1/	$V_{CE(sat)2}$		1.0	V dc
3.	Base emitter voltage (saturated)	3066	Test condition A; I_C = 5 A dc; I_B = 1 A dc (pulsed, see 4.5.1) 1/	$V_{BE(sat)2}$		1.5	V dc
4.	Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 A dc (pulsed, see 4.5.1)	h_{FE2}			
	2N5664, 2N5666				40	120	
	2N5665, 2N5667				25	75	
5.	Collector to emitter cutoff current	3041	See step 1	ΔI_{CES1}		100% of initial value or 20 mA dc, whichever is greater	
6.	Forward-current transfer ratio	3076	See step 4	Δh_{FE2}		±25% change from initial value.	

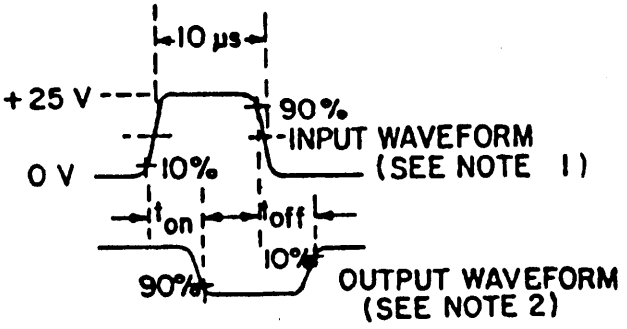
1/ Measured at less than 0.125 inch from case.

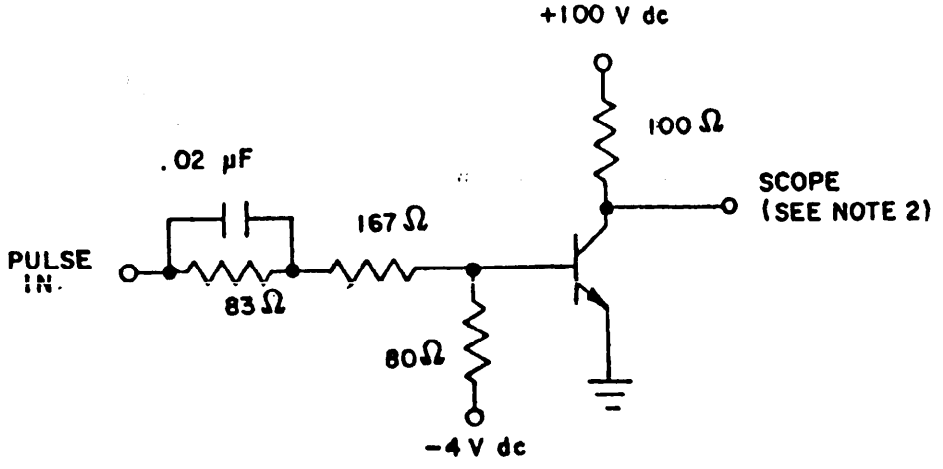




NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\Omega$, PW = 10 µs, duty cycle < 2 %.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r < 15$ ns, $Z_{IN} > 10$ MΩ $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4 V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{G1} = -I_{G2} = 30$ mA and $I_C = 1$ A.

FIGURE 4. Pulse response test circuit for types 2N5664 and 2N5666.





NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\Omega$, PW = 10 µs, duty cycle < 2 %.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{IN} \geq 10$ MΩ $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional bypassing in order to minimize ringing.
5. The input pulse voltages and supply voltages (-4 V dc and +100 V dc) are nominal and shall be adjusted to obtain $I_{B1} = -I_{B2} = 50$ mA and $I_C = 1$ A.

FIGURE 5. Pulse response test circuit for types 2N5665 and 2N5667.

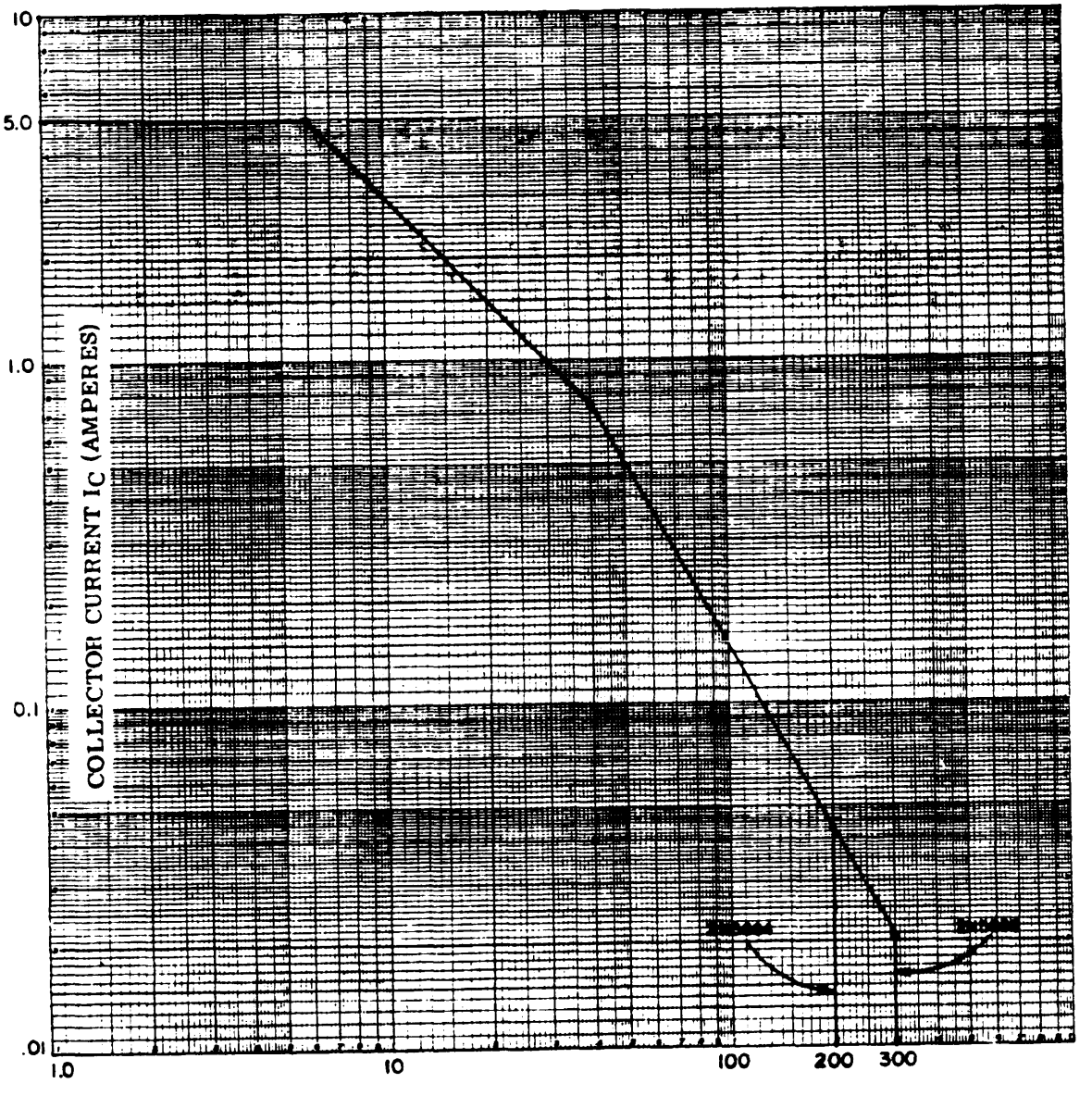


FIGURE 6. Maximum safe operating graph (continuous dc) for types 2N5664 and 2N5666.

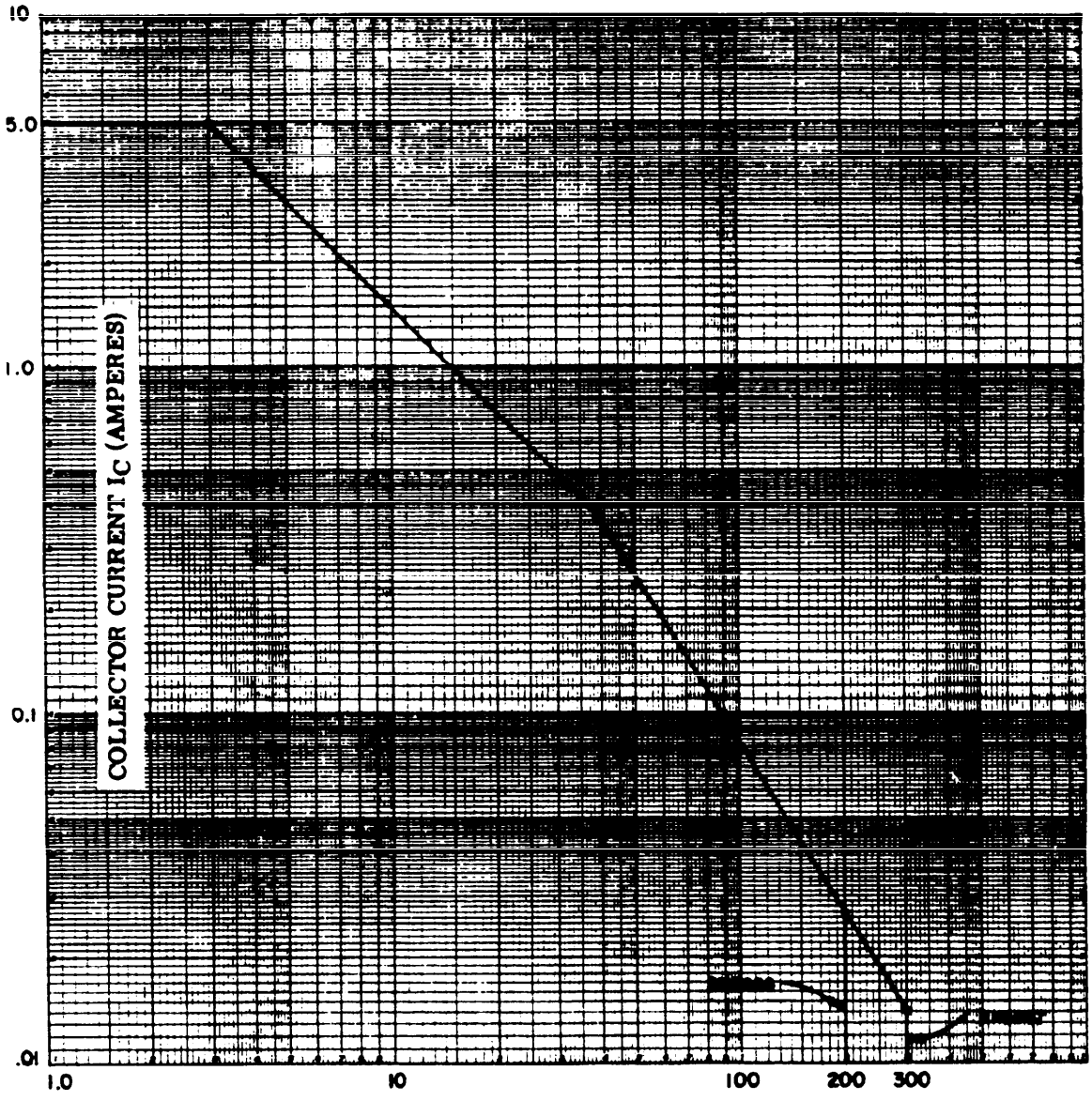


FIGURE 7. Maximum safe operating graph (continuous dc) for types 2N5666 and 2N5667.

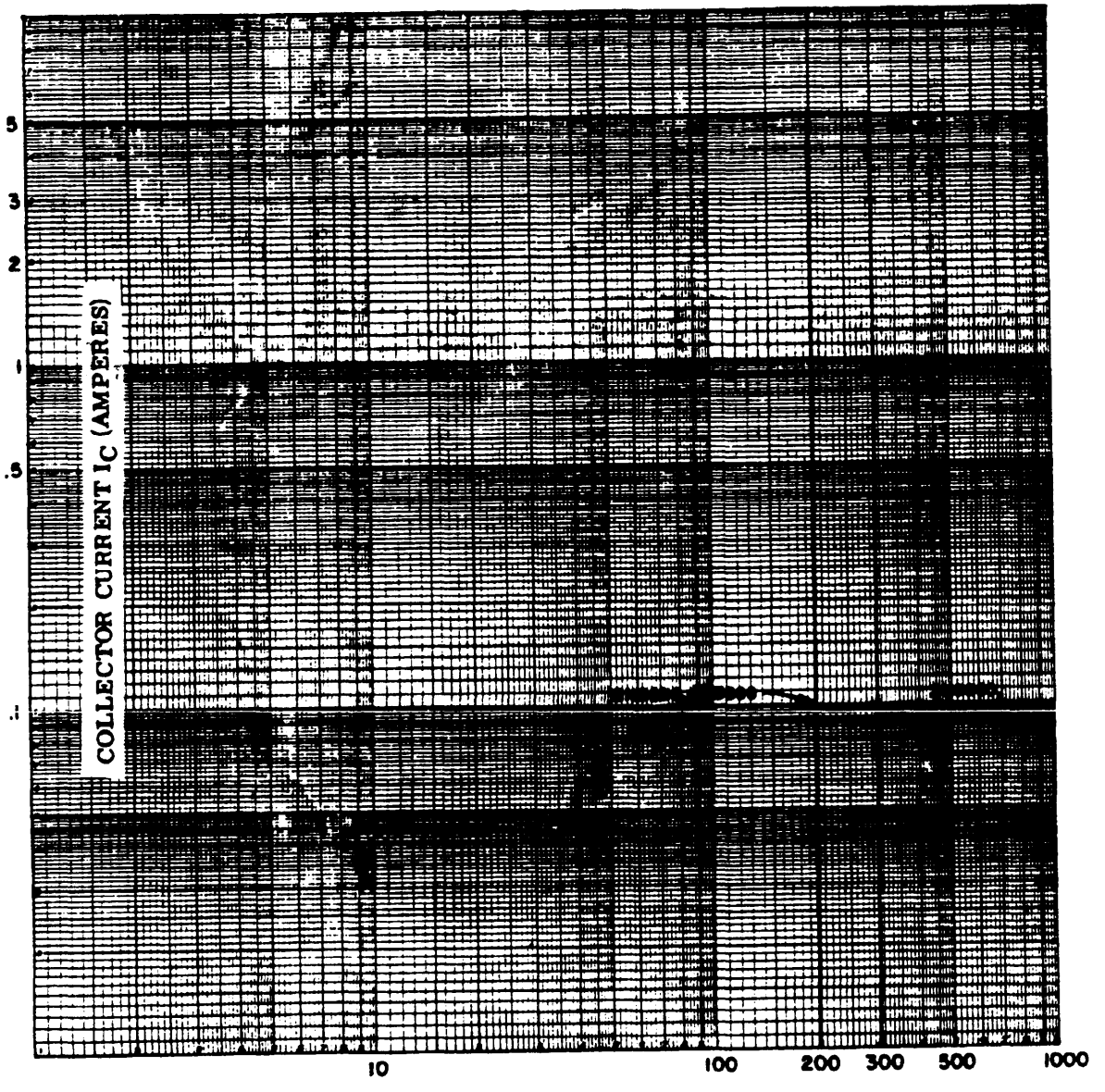


FIGURE 8. Safe operating area for switching between saturation and cutoff (clamped inductive load).

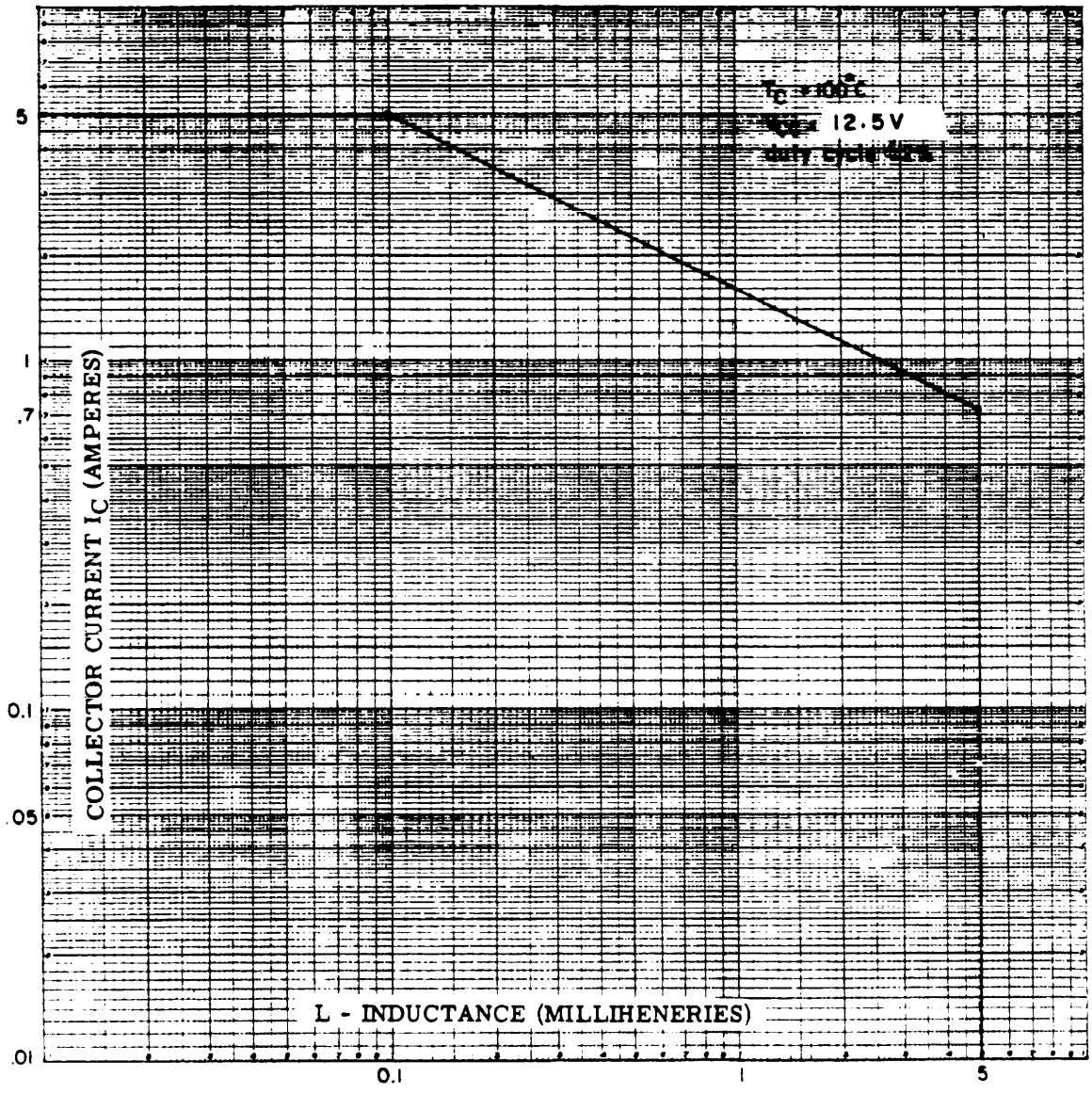


FIGURE 9. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, ER, MI
Navy - SH
Air Force - 11, 19, 80, 85, 99

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1053)